

Modulating p–n transition of two-dimensional perovskites for efficient and stable perovskite/Si tandem photovoltaics

Received: 13 August 2025

Accepted: 2 July 2026

Published online: 30 July 2026

 Check for updates

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The popular and effective top-surface mixed-dimensional heterojunction in n–i–p perovskite photovoltaics is frequently ineffective in p–i–n architectures due to the intrinsic p-type character of most reported two-dimensional (2D) or low-dimensional perovskites. Here we deploy electronic engineering to realize the p- to n-type transition in 2D Ruddlesden–Popper perovskites through molecular dipole tuning and chemically designable n-type defects. Parahalogenated piperidine derivatives induce a favourable p- to n-type transition in 2D perovskites, thereby improving energy-level alignment at the perovskite–electron transport layer interface. This tailored band alignment effectively reduces interfacial energy barriers, suppresses non-radiative recombination losses for wide-bandgap perovskites and enhances stability. Notably, p–i–n wide-bandgap (–1.68 eV) perovskite solar cells with an n-type 2D capping layer exhibit more than 100 mV enhancement in open-circuit voltage, resulting in a certified power conversion efficiency of 33.64% when integrated with industrial 110- μm -thick Czochralski heterojunction silicon. Furthermore, such monolithic perovskite/silicon tandem cells retain 92% of their initial efficiency after 1,100 h of continuous operation under maximum-power-point tracking. Our chemical design of p- to n-type transition for 2D perovskite establishes a molecular-level strategy for optimizing interfacial energetics towards high-performance perovskite/silicon tandem photovoltaics.

Perovskite solar cells (PSCs) have recently achieved power conversion efficiencies (PCEs) approaching 28%¹. This remarkable progress has been largely driven by advances in interface engineering and defect passivation. Among various approaches, constructing two-dimensional (2D) and three-dimensional (3D) heterojunctions offers an effective strategy for mitigating interfacial losses, passivating defects and boosting device performance. These heterojunctions also play a vital role in a range of wide-bandgap (WBG) perovskite-based

tandem configurations, including perovskite/perovskite^{2,3}, perovskite/organic⁴, perovskite/copper indium gallium selenide⁵ and perovskite/silicon^{6–10}. In 2D/3D heterojunctions, the deposited 2D perovskites are able to interact with undercoordinated Pb ions and halide vacancies at the surface or grain boundaries of the 3D perovskites, inhibiting non-radiative recombination by neutralizing these defect states¹¹. Additionally, thanks to their layered hydrophobic architecture, 2D perovskites offer a robust physical barrier against environmental

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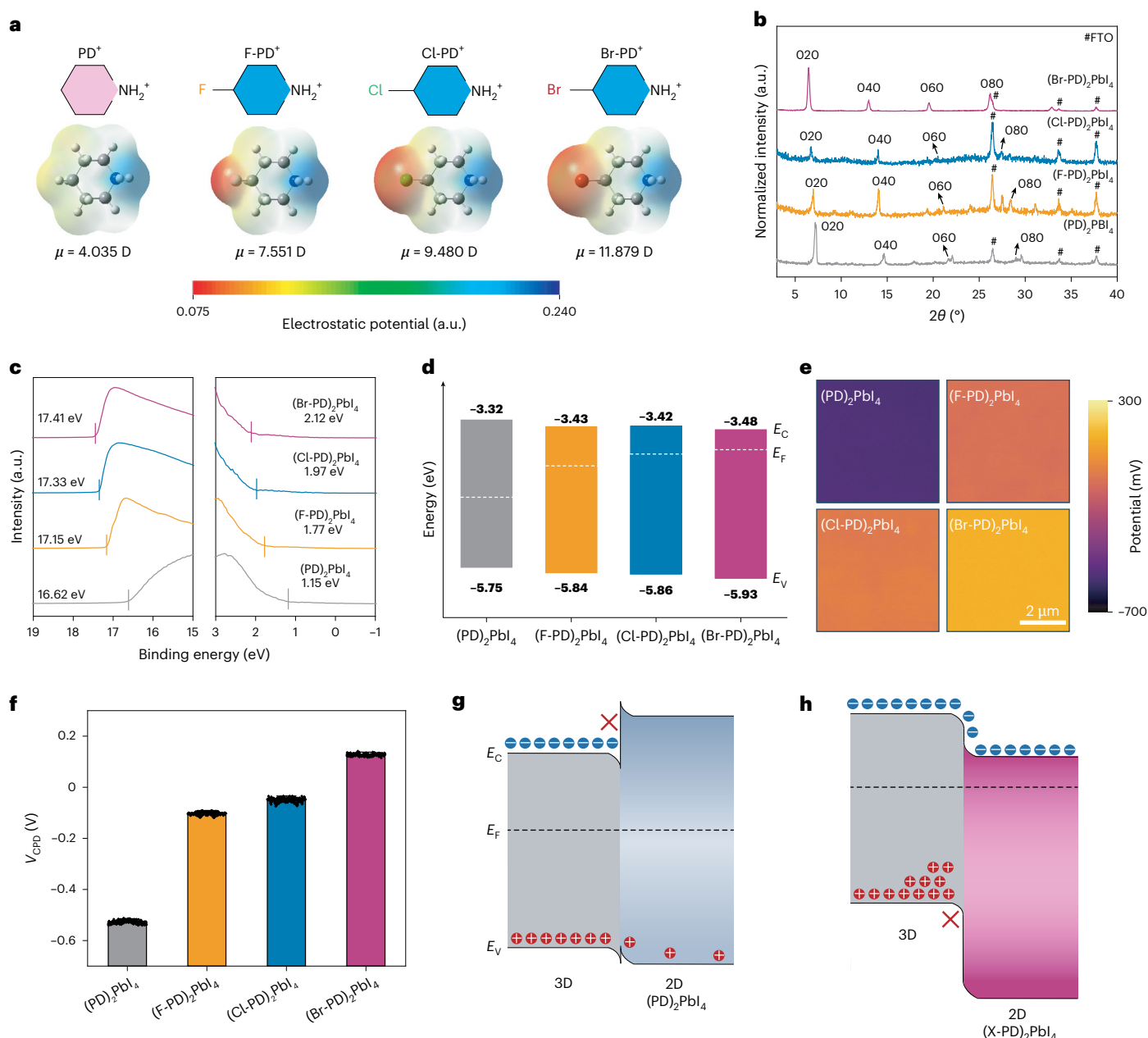


Fig. 1 | Electronic properties of 2D RP perovskites with different organic spacers. a, Chemical structures of the synthesized molecules, showing their electrostatic potentials and dipole moments. **b**, XRD patterns of the 2D thin films. # denotes the diffraction peaks from the FTO glass. θ represents the diffraction angle. **c**, UPS spectra, including the photoemission cutoff energy and valence-band region. **d**, Energy-level diagrams derived from the UPS and UV-vis absorbance spectra. **e**, KPFM images with a standardized range of 1 V. **f**, Surface

potential obtained from the contact potential difference distributions. For each sample, 256 values ($n = 256$) were extracted from the corresponding KPFM image. Data are presented as mean values, with error bars indicating the standard error of the mean. **g, h**, Schematic diagrams of the band alignment in the resulting 2D/3D heterojunctions based on pristine $(PD)_2PbI_4$ (**g**) and halogen substituted PD $(X-PD)_2PbI_4$, X = F, Cl, Br (**h**).

stressors of moisture and oxygen, contributing to enhanced device stability^{12,13}. Despite their success in n-i-p PSCs with normal-bandgap absorbers, state-of-the-art 2D/3D heterojunctions face substantial challenges in inverted p-i-n architectures, particularly for WBG devices, where forming an energetically favourable interface that both enables selective charge extraction and suppresses interfacial non-radiative recombination remains elusive¹⁴. This is because most 2D perovskites tend to exhibit a more p-type nature than their 3D counterparts^{15,16}, resulting in unfavourable energy alignment with n-type electron transport layers used in p-i-n devices. To circumvent this issue, further development of 2D/3D perovskite heterojunctions should prioritize

electronic-structure modulation for efficient defect passivation and charge transport.

In this study, we develop 2D/3D heterojunctions by molecular dipole engineering via incorporating polar substituents or asymmetric spacers^{17–19}. Tuning the dipole moments of the spacers enables continuous control over the electronic properties of 2D Ruddlesden–Popper (RP) perovskites, driving a transition from p- to n-type behaviour. Among the investigated spacers, the bromide piperidine (PD)-based 2D RP perovskite $(Br-PD)_2PbI_4$ exhibits the most pronounced n-type character. Integrated with 3D WBG perovskite, this n-type 2D layer forms a favourable heterojunction that simultaneously passivates

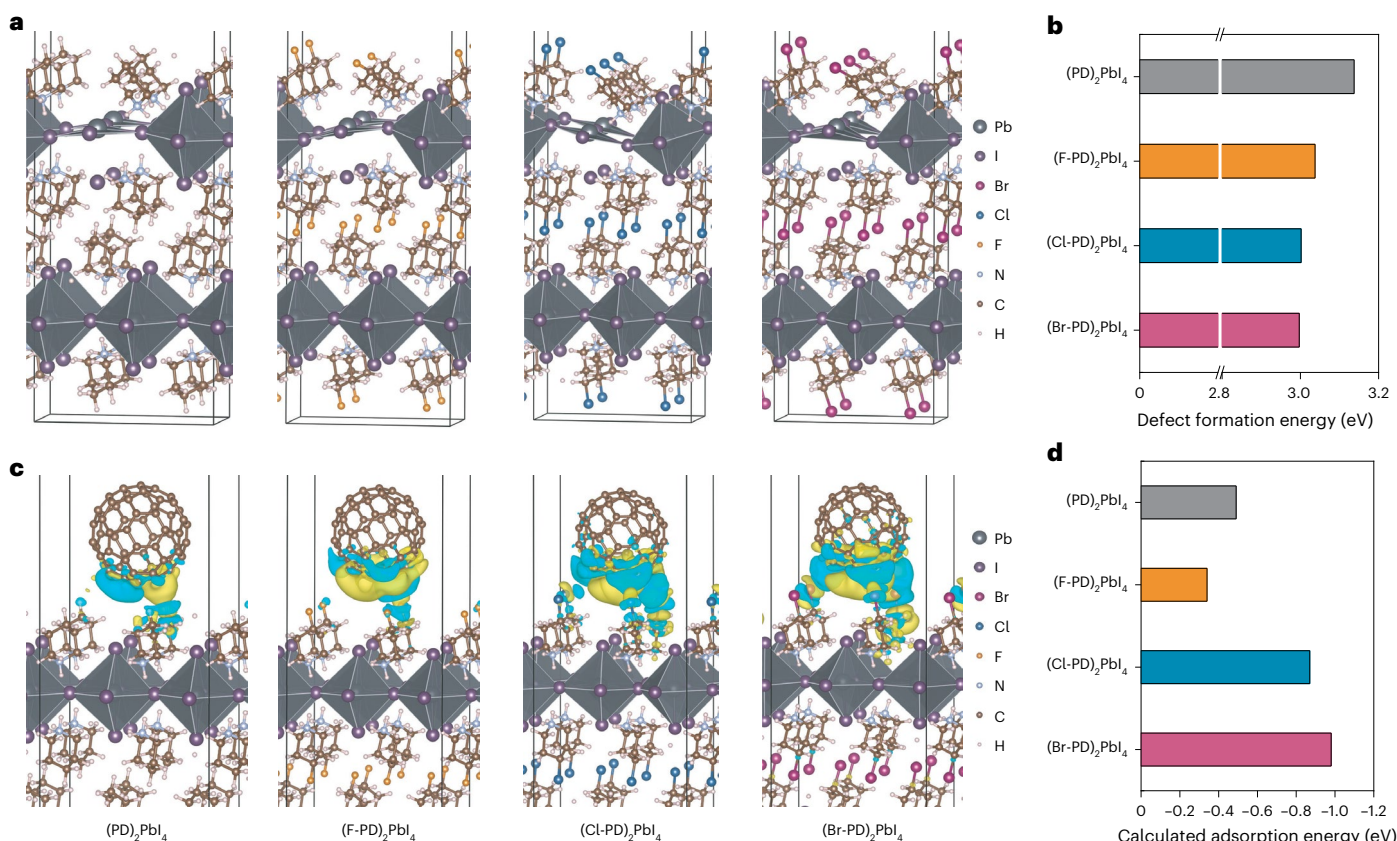


Fig. 2 | DFT calculations of different 2D RP perovskites and their interfaces with C₆₀. **a**, Optimized structures of 2D perovskite surfaces containing iodide vacancies located at the outermost atomic layer adjacent to the vacuum region. **b**, Calculated defect formation energies under various organic spacer

conditions. **c**, 3D charge-density differences at C₆₀-2D perovskite interfaces (isosurface = $3.08 \times 10^{-5} \text{ e bohr}^{-3}$). Blue and yellow denote electron depletion and accumulation, respectively. **d**, Calculated adsorption energies of C₆₀ on 2D perovskite surfaces with different organic spacers.

defects and optimizes energy-level alignment, thereby promoting charge transport and suppressing non-radiative recombination. Such electronic tunability increases the open-circuit voltage (V_{OC}) of WBG PSCs (bandgap: 1.68 eV) from 1.16 V to 1.26 V and the PCE from 20.19% to 23.40%. Furthermore, perovskite/silicon tandem cells achieve a certified champion PCE of 33.64% with negligible hysteresis. Owing to effective suppression of phase segregation, both devices exhibit excellent operational stability, retaining 96% of their initial PCE after 1,000 hours of maximum-power-point tracking (MPPT) for single-junction PSCs and 92% after 1,100 hours for tandem cells.

Results and Discussion

Molecular design for tunable 2D perovskites

We designed a series of passivating ligands based on the PD scaffold with systematic halogen substitution. The incorporation of halogen atoms influences the electron distribution within the molecule (Fig. 1a), generating electron-rich and electron-deficient regions. This leads to an asymmetrical electrostatic potential and a molecular dipole moment that increases with halogen radius, probably due to the higher polarizability and more diffuse electron density of larger halogens²⁰. Notably, the Br-PD⁺ exhibits the highest dipole moment of 11.879 D, nearly three times larger than that of the unmodified PD⁺ (4.053 D). The varying molecular dipole moments could affect the electronic properties¹⁹. To investigate this effect, we prepared 2D RP perovskite films ((X-PD)₂PbI₄, X = F, Cl, Br), the characteristic diffraction peaks of which are confirmed in Fig. 1b. For pristine (PD)₂PbI₄ film, the Fermi level (E_F) was determined to be -4.60 eV, which is closer to the valence-band maximum (E_V) of -5.75 eV (Fig. 1c and Supplementary Figs. 1 and 2). With halogenated organic spacers, the E_F values of (F-PD)₂PbI₄, (Cl-PD)₂PbI₄

and (Br-PD)₂PbI₄ are -4.07 eV, -3.89 eV and -3.81 eV, respectively. The increasing E_F approaching the edge of conduction band minimum (E_C) suggests more n-type characteristics (Fig. 1d). To verify these uniformities, Kelvin probe force microscopy (KPFM) was performed on larger areas (Fig. 1e and Supplementary Fig. 3); the work function displacements are shown in Fig. 1f. The average contact potential difference gradually increases from -0.54 V to 0.14 V, consistent with the E_F shifting in ultraviolet photoelectron spectroscopy (UPS) spectra. Hence, the halogenated strategy of organic spacers can regulate molecular dipoles and further affect electronic properties of the formed 2D perovskites, leading to a p- to n-type transition.

When in contact with 3D perovskite (Fig. 1g and Supplementary Figs. 4 and 5), a type I heterojunction was formed for non-substituted (PD)₂PbI₄, with a substantial band offset at E_C (-1.0 eV) and a small offset at E_V (-0.2 eV). Modified by halogenated 2D perovskites, the downshifted E_C can facilitate electron transfer and enhance their extraction to the electron transport layer (that is, a type II band alignment in Fig. 1h). Meanwhile, the deeper E_V induced a large ΔE_V (valence band offset between 2D and 3D perovskites), making 2D perovskites a hole-blocking layer that could mitigate undesired charge recombination. These structural advantages were further demonstrated by the improved intensities in steady-state photoluminescence (PL) and decay times in time-resolved PL measurements (Supplementary Fig. 6 and Supplementary Table 1).

The electronic properties of perovskite are generally determined by the presence and nature of intrinsic defects^{18,21,22}. We performed first-principle calculations based on density functional theory (DFT; Supplementary Fig. 7), focusing on a representative n-type defect of iodide vacancies (V_I) in various 2D perovskite systems (Fig. 2a).

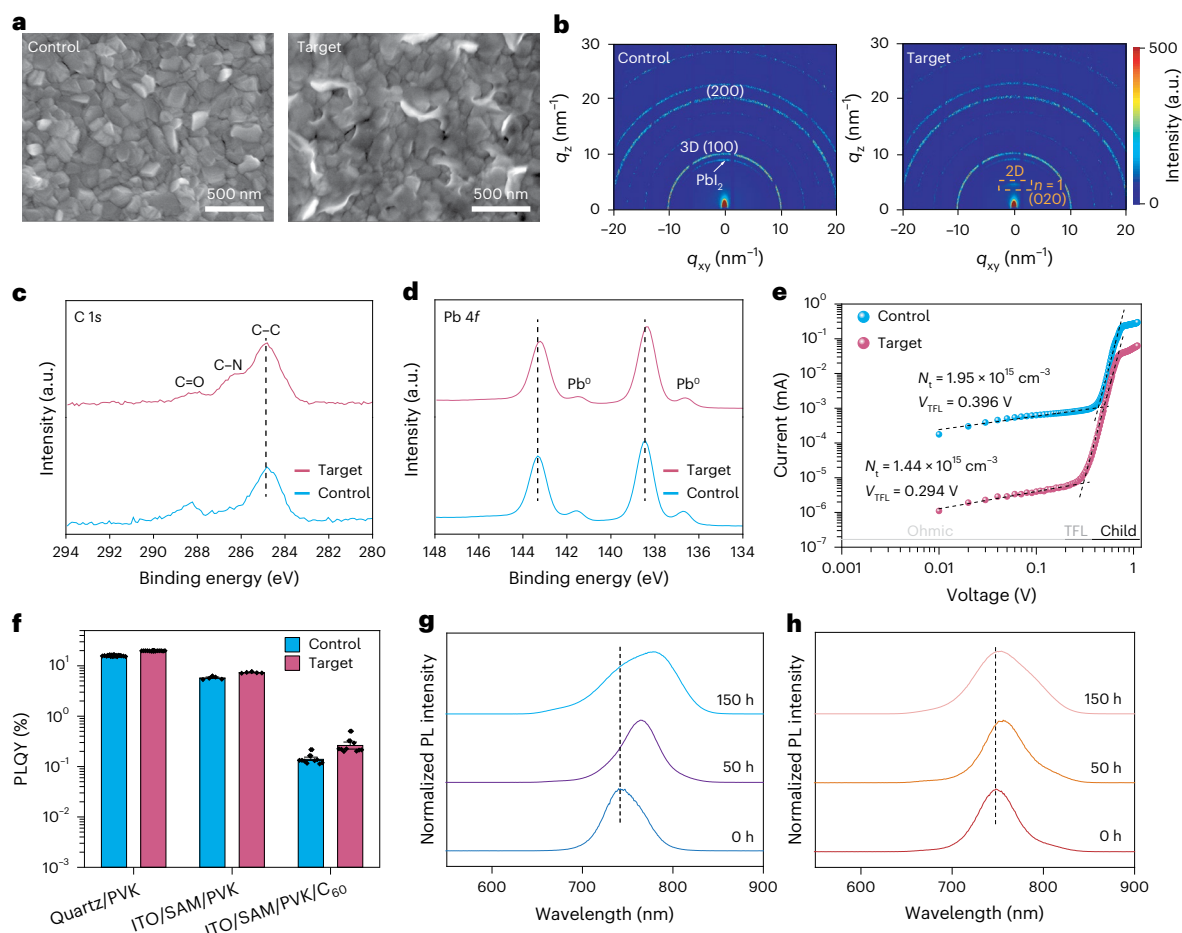


Fig. 3 | Properties of 3D perovskite films without (control) or with (target) Br-PDI treatment. **a**, Top-view SEM images. **b**, 2D GIWAXS patterns collected at an incidence angle of 0.1° . **c, d**, High-resolution XPS of C 1s (**c**) and Pb 4f (**d**) core-level spectra. **e**, Space charge-limited current measurements of electron-only devices (ITO/SnO₂/perovskite/(without or with Br-PDI)/C₆₀/BCP/Ag). **f**, PLQY measured in air without any encapsulation. For each sample, data from independent

measurements (black squares) are presented as mean values, with error bars indicating the standard error of the mean. **g, h**, Evolution of PL spectra ageing under continuous light soaking (100 mW cm^{-2} , white LED) in a N₂ atmosphere for control (**g**) and target (**h**). Note that all the perovskite films were deposited on MeO-2PACz-coated ITO substrates unless otherwise specified. TFL, trap-filled limit voltage; PVK, perovskite; SAM, self-assembled materials.

As summarized in Fig. 2b, the halogen spacers reduce the defect formation energies (E_f), with the lowest E_f values shown in (Br-PD)₂PbI₄. This confirmed the pronounced n-type character induced by the Br-PD⁺ spacer, consistent with our UPS and KPFM measurements. Organic spacers with larger molecular dipoles can generate a strong local electrostatic field¹⁷. This modifies the electronic density around the Pb-I framework and stabilizes the positive electrostatic environment created on removal of an iodide, thereby reducing the formation energy of iodide vacancies through the vacancy–dipole interaction (also supported by Supplementary Fig. 8 and Supplementary Table 2)²³.

Additionally, Fig. 1g, h suggests that molecular dipole engineering contributes to energy-level management that could be beneficial for electron transfer from perovskite to the adjacent electron transport layer. To further examine this interface, we investigated the interactions between C₆₀ and 2D perovskites. As illustrated in Fig. 2c, the charge-density difference shows that the (Br-PD)₂PbI₄ exhibits the strongest charge-transfer tendency with C₆₀, followed by (Cl-PD)₂PbI₄, whereas (PD)₂PbI₄ and (F-PD)₂PbI₄ induce only moderate charge redistribution within C₆₀ (ref. 24). The corresponding adsorption energies (E_{ads}) are -0.49 eV , -0.34 eV , -0.87 eV and -0.98 eV for (PD)₂PbI₄, (F-PD)₂PbI₄, (Cl-PD)₂PbI₄ and (Br-PD)₂PbI₄, respectively. The weak adsorption on (F-PD)₂PbI₄ can be attributed to the low polarizabilities of fluorine, which suppresses the van der Waals interactions, and its strong electronegativity, which reduces the surface electron-donating

ability towards Lewis acid C₆₀ (ref. 25). Among these halide substitutions, the larger and more polarizable Br atom generates a more pronounced σ hole along the C–Br bond direction. This positively charged region can interact with the electron-rich π cloud of C₆₀, resulting in a stronger halogen– π interaction compared with other counterparts²⁶. Hence, (Br-PD)₂PbI₄ exhibits the strongest interaction with C₆₀ (Fig. 2d), which helps reduce the interfacial energetic mismatch and facilitate efficient charge transfer¹⁷.

Properties of 2D/3D perovskite films

The p-to-n transition observed in 2D RP perovskites is reflected in the surface electronic structure of 3D perovskites (Supplementary Fig. 9 and Supplementary Table 3). Because 2D films exhibit comparable layer numbers (Supplementary Fig. 10), the improved interfacial energetics are attributed to the spacer-induced dipole moment. Given its favourable energy-level alignment, Br-PDI was chosen to treat the 3D WBG perovskite film (referred to as the ‘target’). During annealing (Supplementary Fig. 11), the initial intensity increase indicated the formed 2D perovskite²⁷. Br-PDI fully covered the surface and reacted with excess PbI₂ (Fig. 3a and Supplementary Fig. 12)²⁸, leading to a smoother surface (Supplementary Fig. 13). Because the ground grain boundaries are considered valleys on the surface and typically act as recombination reservoirs²⁹, the formed 2D capping layer could passivate these grain boundaries to reduce

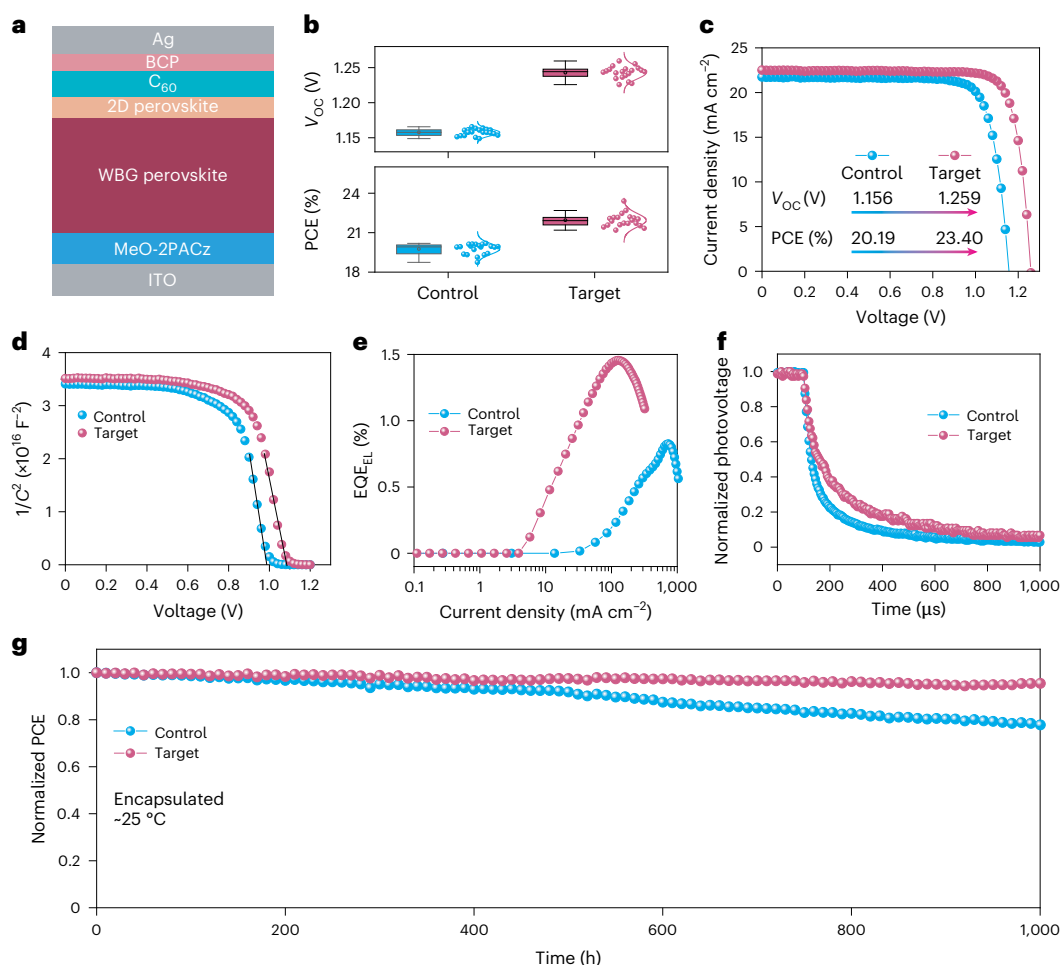


Fig. 4 | Photovoltaic performance of single-junction WBG PSCs. a, Schematic of the device structure. **b**, Statistical comparison of the V_{oc} and PCEs. Each group comprises 20 independent devices. The box bounds represent the 25th and 75th percentiles, the centre line indicates the median, and the black square denotes the mean value. **c**, Champion $J-V$ curves from the reverse scan. **d**, Mott-Schottky plots obtained from $C-V$ measurements. **e**, EQE_{EL} measurements under current

densities from 0.1 mA cm⁻² to 1,000 mA cm⁻². **f**, Transient photovoltage spectra of the control device (without the 2D layer) and the target device (with the incorporated 2D perovskite layer). **g**, Long-term operational stability under MPPT conditions, with data recorded every 10 h during continuous light soaking (100 mW cm⁻², white LED) in a N₂ atmosphere.

the probability of recombination. The schematic formation processing of Br-PDI-induced 2D/3D heterostructures is illustrated in Supplementary Fig. 14. The Br-PDI-treated film has a new peak at 6.5° and a suppressed PbI₂ intensity at 12.8° in X-ray diffraction (XRD) patterns (Supplementary Fig. 15). Grazing incidence wide-angle X-ray scattering (GIWAXS) further confirmed the formation of 2D RP perovskites, together with enhanced diffraction intensity and weakened PbI₂ patterns in the target perovskite film (Fig. 3b and Supplementary Fig. 16).

The formed 2D phase can reconfigure the electronic and chemical states of the perovskite films. Additional peaks at 286.3 eV (Fig. 3c) and 402.0 eV (Supplementary Fig. 17) confirmed the presence of 2D perovskite on the surface. The target film shows a downward shift of the Pb 4*f* orbital (Fig. 3d), corroborating the interfacial interactions induced by the 2D layer³⁰. Besides these main peaks at 138.3 eV (Pb 4*f*_{7/2}) and 143.2 eV (Pb 4*f*_{5/2}), the other two small peaks at 136.6 eV and 141.4 eV are ascribed to metallic Pb (Pb⁰) (ref. 31), which could induce high-density defects in the perovskite film. By contrast, the formed 2D perovskite can mitigate unsaturated Pb, with the peak area ratio of Pb⁰/Pb 4*f* decreasing from 12.7% to 9.3%. Therefore, the X-ray photoelectron spectroscopy (XPS) analysis of the elemental composition suggests that the 2D capping layer can achieve more effective interaction and defect passivation on 3D perovskite surfaces.

The target film showed a lower trap density (N_t) than the control film (1.44×10^{15} cm⁻³ versus 1.95×10^{15} cm⁻³; Fig. 3e and Supplementary Fig. 18). Large-area PL mappings (100 × 100 μm²) revealed markedly enhanced and more uniform emission intensity in the target film (Supplementary Fig. 19). Photoluminescence quantum yield (PLQY; Fig. 3f) indicated suppressed recombination losses, particularly at the perovskite-C₆₀ heterointerface. After 150 h of light soaking (Fig. 3g,h), the control film showed a notable PL peak shifting from 745 nm to 780 nm, whereas the Br-PDI-treated film remained stable (for example, with only 4-nm shifts). The PbI₂ peak became markedly more pronounced in the control film than that in the Br-PDI-treated perovskite film (Supplementary Fig. 20). These findings demonstrate that the 2D capping layer notably suppressed phase segregation for the underlying 3D WBG perovskite, enhancing both light and thermal stability. The improved stability originates from simultaneous defect passivation and inhibited ion migration (Supplementary Fig. 21).

Optoelectronic performance of PSCs

Single-junction WBG PSCs were first fabricated as shown in Fig. 4a. Transmission electron microscopy (Supplementary Fig. 22) confirmed that a thin 2D layer (~10 nm) had formed on the WBG perovskite surface with an interplanar spacing ($d_{spacing}$) of 1.363 nm (Supplementary Fig. 23), consistent with the (020) plane of the 2D RP

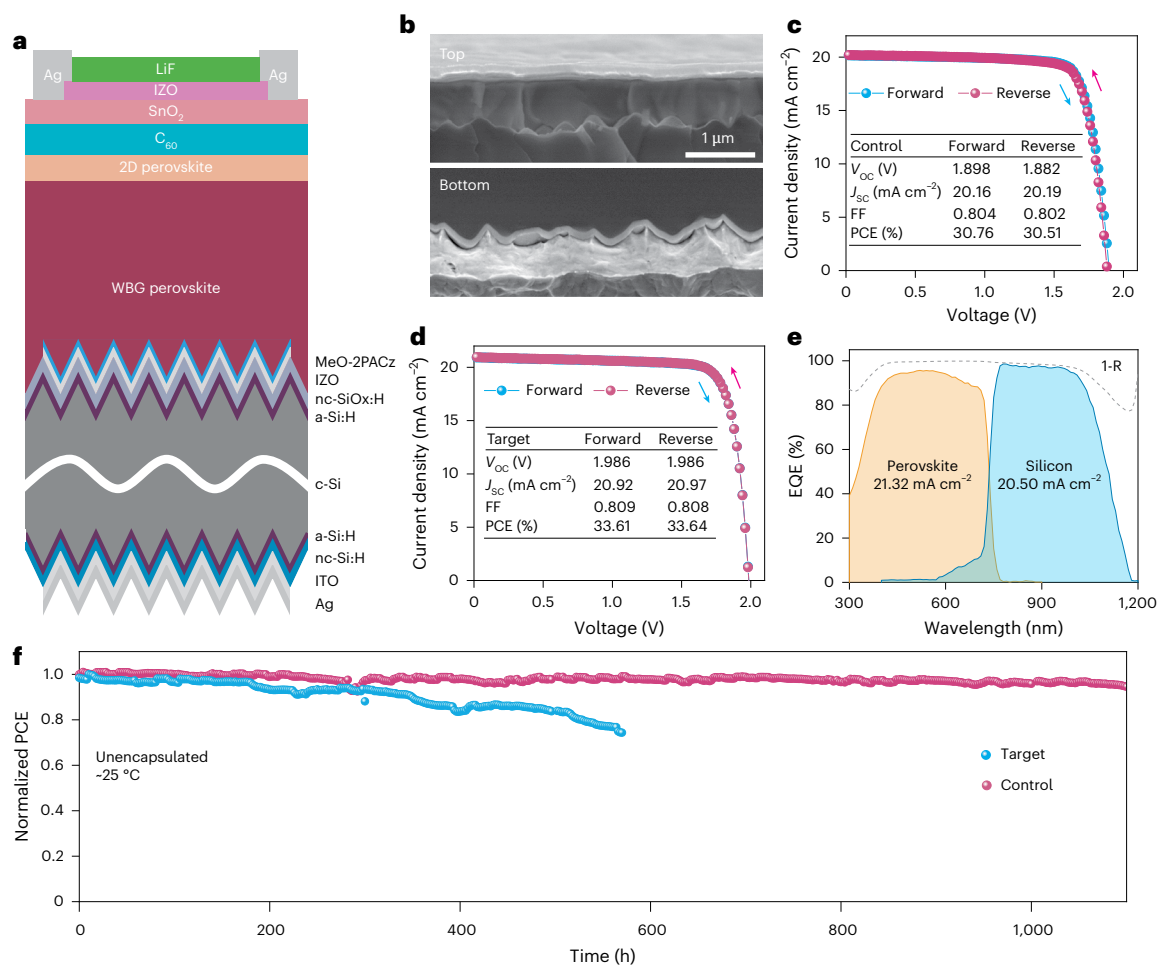


Fig. 5 | Photovoltaic performance of monolithic perovskite/Si tandem cells.

a, Schematic of the device structure, including transparent electrode made with tin oxide (SnO₂), IZO, silver grid (Ag) and antireflective lithium fluoride (LiF). **b**, Cross-sectional SEM images of the top and bottom cells. **c,d**, Comparison of champion *J*-*V* curves for the control (**c**) and target (**d**) devices, including forward

and reverse scans, together with the extracted device parameters. All data shown in **d** are certified values. **e**, EQE spectra and reflection (denoted as 1-R) measured in-house. **f**, Long-term operational stability under MPPT conditions, with data recorded every 2 h during continuous light soaking (100 mW cm⁻², white LED) in a N₂ atmosphere.

perovskite (Br-PD)₂PbI₄ (*n* = 1). This ultrathin layer promotes electron transfer at the perovskite-C₆₀ interface (Supplementary Fig. 24 and Supplementary Tables 4 and 5) and effectively enhances photovoltaic performance (Supplementary Fig. 25). The target devices exhibited enhanced open-circuit voltage (*V*_{oc}) from 1.16 ± 0.01 V to 1.24 ± 0.01 V (arithmetic mean ± standard deviation), contributing to average PCEs from 19.76 ± 0.40% to 21.94 ± 0.51% (Fig. 4b). The best-performing device achieved a *V*_{oc} of 1.259 V, a short-circuit current density (*J*_{sc}) of 22.51 mA cm⁻², a fill factor (FF) of 0.825 and a PCE of 23.40% (Fig. 4c, Supplementary Fig. 26 and Supplementary Table 6), with a closely stabilized PCE of 23.15% (Supplementary Fig. 27) and certified PCE of 23.38% (Supplementary Fig. 28). The external quantum efficiency (EQE) spectra showed an integrated *J*_{sc} of 21.92 mA cm⁻² and an unchanged electronic bandgap (Supplementary Fig. 29).

Mott-Schottky plots from the capacitance-voltage (*C*-*V*) measurements showed that the built-in voltage (*V*_{bi}) increased from 0.99 V and 1.09 V (Fig. 4d). The target device with a reduced ideality factor (*n*_{id}, 1.45 versus 1.73; Supplementary Fig. 30) suggested the mitigated trap-assisted recombination³². The external electroluminescence quantum efficiency (EQE_{EL}) of the target device was 0.79% (Fig. 4e), leading to a reduced *V*_{oc} loss caused by non-radiative recombination (Δ*V*_{oc,nrad}) to 125.1 mV. By contrast, the control device showed only 0.01% EQE_{EL} with an increased Δ*V*_{oc,nrad} of 238.1 mV. Further transient photovoltage measurements exhibited an extended photovoltage decay time

in the target device, approaching a threefold increase from 42.8 μs to 126.1 μs (Fig. 4f). Similar trends were observed in electrochemical impedance spectroscopy, where Nyquist plots showed reduced series resistance (from 26.92 Ω to 15.46 Ω) and increased charge recombination resistance (from 3,400 Ω to 5,601 Ω) for the target devices (Supplementary Figs. 31 and 32 and Supplementary Table 7). These results indicate enhanced charge transport and suppressed recombination enabled by our interface engineering strategy. Therefore, the formed 2D/3D heterojunction can effectively suppress non-radiative recombination processes throughout the complete device.

Effective suppression of non-radiative recombination not only enhances efficiency but also improves the stability of PSCs³³. Long-term stability was evaluated following International Summit on Organic Photovoltaic Stability (ISOS) protocols³⁴. For shelf-life assessment (equivalent to ISOS-D-2 in Supplementary Fig. 33), the unencapsulated target device maintained 87% of its initial performance after 800 h, superior to the control device, which dropped to less than 61% of the initial PCE. Furthermore, operational stability based on MPPT was also monitored (equivalent to ISOS-L-II). After 1,000 h, the encapsulated target device maintained 96% of its initial efficiency, whereas the control device retained only 78% (Fig. 4g and Supplementary Fig. 34). At elevated temperature (for example, -65 °C, equivalent to ISOS-L-2I), the encapsulated Br-PDI-based device retained 90% of its initial PCE after 500 h, whereas the control device degraded to 66% (Supplementary Fig. 35).

Due to the well-suited processing and integration with Si, our approach was further extended to perovskite/Si tandem solar cells. Incorporating with our rationally designed 2D/3D heterojunction, two-terminal monolithic perovskite/Si tandem cells were fabricated (Fig. 5a,b). Compared with the control devices (Supplementary Fig. 36), the target tandem devices exhibit improved average V_{oc} (1.88 ± 0.02 V to 1.98 ± 0.01 V) and FF (0.78 ± 0.02 to 0.81 ± 0.01). The champion target device showed negligible hysteresis between forward and reverse current density–voltage (J – V) scans (Fig. 5c,d). Performance was independently certified by the National Center of Inspection on Solar Photovoltaic Products Quality (Supplementary Fig. 37), yielding a champion PCE of 33.64%. The integrated J_{sc} values of the two subcells calculated from the EQE spectra were 21.32 mA cm^{-2} and 20.50 mA cm^{-2} , respectively (Fig. 5e). The certified EQE exhibits consistent spectral features and integrated J_{sc} (Supplementary Fig. 38). For long-term operation (equivalent to ISOS-L-II; Supplementary Fig. 39), the tandem cell still maintained more than 92% of its initial efficiency after 1,100 h. By contrast, the control device maintained only 80% of its initial efficiency after 520 h (Fig. 5f). Hence, enhanced operational stability demonstrates the robustness of the modified interface. Taken together, both single and tandem cells achieved the best performance among the heterojunction engineering strategies explored (Supplementary Table 8), clearly demonstrating the efficacy of our approach.

Online content

Any methods, additional references, Nature Portfolio reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at <https://doi.org/10.1038/s41566-026-01979-z>.

References

- Best Research-Cell Efficiencies (NLR, 2026); <https://www.nlr.gov/media/docs/libraries/pv/cell-pv-eff.pdf>
- Wang, Y. et al. Homogenized contact in all-perovskite tandems using tailored 2D perovskite. *Nature* **635**, 867–873 (2024).
- Liu, Z. et al. All-perovskite tandem solar cells achieving >29% efficiency with improved (100) orientation in wide-bandgap perovskites. *Nat. Mater.* **24**, 252–259 (2025).
- Zhang, Z. et al. Suppression of phase segregation in wide-bandgap perovskites with thiocyanate ions for perovskite/organic tandems with 25.06% efficiency. *Nat. Energy* **9**, 592–601 (2024).
- Gharibzadeh, S. et al. 2D/3D heterostructure for semitransparent perovskite solar cells with engineered bandgap enables efficiencies exceeding 25% in four-terminal tandems with silicon and CIGS. *Adv. Funct. Mater.* **30**, 1909919 (2020).
- Chen, Y. et al. Nuclei engineering for even halide distribution in stable perovskite/silicon tandem solar cells. *Science* **385**, 554–560 (2024).
- Ugur, E. et al. Enhanced cation interaction in perovskites for efficient tandem solar cells with silicon. *Science* **385**, 533–538 (2024).
- Liu, J. et al. Perovskite/silicon tandem solar cells with bilayer interface passivation. *Nature* **635**, 596–603 (2024).
- Kim, D. et al. Efficient, stable silicon tandem cells enabled by anion-engineered wide-bandgap perovskites. *Science* **638**, 155–160 (2020).
- Pei, F. et al. A binary 2D perovskite passivation for efficient and stable perovskite/silicon tandem solar cells. *Nat. Commun.* **15**, 7024 (2024).
- Gharibzadeh, S. et al. Record open-circuit voltage wide-bandgap perovskite solar cells utilizing 2D/3D perovskite heterostructure. *Adv. Energy Mater.* **9**, 1803699 (2019).
- Wang, Z. et al. Efficient ambient-air-stable solar cells with 2D-3D heterostructured butylammonium-caesium-formamidinium lead halide perovskites. *Nat. Energy* **2**, 1–10 (2017).
- Randi, A. et al. Damp heat-stable perovskite solar cells with tailored-dimensionality 2D/3D heterojunctions. *Science* **376**, 73–77 (2022).
- Xiao, Y., Yang, X., Zhu, R. & Snaith, H. J. Unlocking interfaces in photovoltaics. *Science* **384**, 846–848 (2024).
- Sidhik, S. et al. Deterministic fabrication of 3D/2D perovskite bilayer stacks for durable and efficient solar cells. *Science* **377**, 1425–1430 (2022).
- Ma, K. et al. Holistic energy landscape management in 2D/3D heterojunction via molecular engineering for efficient perovskite solar cells. *Sci. Adv.* **9**, eadg0032 (2023).
- Li, T. et al. Inorganic wide-bandgap perovskite subcells with dipole bridge for all-perovskite tandems. *Nat. Energy* **8**, 610–620 (2023).
- Jiang, X. et al. Surface heterojunction based on n-type low-dimensional perovskite film for highly efficient perovskite tandem solar cells. *Natl Sci. Rev.* **11**, nwa055 (2024).
- Wang, G. et al. Reducing voltage loss via dipole tuning for electron-transport in efficient and stable perovskite-silicon tandem solar cells. *Adv. Energy Mater.* **14**, 2401029 (2024).
- Cavallo, G. et al. The halogen bond. *Chem. Rev.* **116**, 2478–2601 (2016).
- Jiang, Q. et al. Surface reaction for efficient and stable inverted perovskite solar cells. *Nature* **611**, 278–283 (2022).
- Xiong, W. et al. Controllable p- and n-type behaviours in emissive perovskite semiconductors. *Nature* **633**, 344–350 (2024).
- Pazoki, M., Wolf, M. J., Edvinsson, T. & Kullgren, J. Vacancy dipole interactions and the correlation with monovalent cation dependent ion movement in lead halide perovskite solar cell materials. *Nano Energy* **38**, 537–543 (2017).
- Guo, H. et al. Immobilizing surface halide in perovskite solar cells via calix[4]pyrrole. *Adv. Mater.* **35**, 2301871 (2023).
- Shao, Y., Xiao, Z., Bi, C., Yuan, Y. & Huang, J. Origin and elimination of photocurrent hysteresis by fullerene passivation in $\text{CH}_3\text{NH}_3\text{PbI}_3$ planar heterojunction solar cells. *Nat. Commun.* **5**, 1–7 (2014).
- Kanao, E. et al. Separation of halogenated benzenes enabled by investigation of halogen– π interactions with carbon materials. *Chem. Sci.* **11**, 409–418 (2020).
- Wang, T. et al. Dimensional regulation from 1D/3D to 2D/3D of perovskite interfaces for stable inverted perovskite solar cells. *J. Am. Chem. Soc.* **146**, 7555–7564 (2024).
- Luo, D. et al. Enhanced photovoltage for inverted planar heterojunction perovskite solar cells. *Science* **360**, 1442–1446 (2018).
- Jiang, Q. et al. Surface passivation of perovskite film for efficient solar cells. *Nat. Photonics* **13**, 460–466 (2019).
- Xue, J. et al. Reconfiguring the band-edge states of photovoltaic perovskites by conjugated organic cations. *Science* **371**, 636–640 (2021).
- Zhang, W. et al. Enhanced optoelectronic quality of perovskite thin films with hypophosphorous acid for planar heterojunction solar cells. *Nat. Commun.* **6**, 1–9 (2015).
- Li, B. et al. Suppressing interfacial recombination with a strong-interaction surface modulator for efficient inverted perovskite solar cells. *Adv. Energy Mater.* **12**, 2202868 (2022).
- Peng, W. et al. Reducing nonradiative recombination in perovskite solar cells with a porous insulator contact. *Science* **379**, 683–690 (2023).
- Khenkin, M. V. et al. Consensus statement for stability assessment and reporting for perovskite photovoltaics based on ISOS procedures. *Nat. Energy* **5**, 35–49 (2020).

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Methods

Materials

Formamidinium iodide was synthesized using formamidine acetate (99%) and hydroiodic acid (HI, 57% in water, with 1.5% H_3PO_2) from Merck Co., Ltd. Lead iodide (PbI_2 , 99.99%), lead bromide (PbBr_2 , 99.99%), lead chloride (PbCl_2 , 99.99%) and [2-(3,6-dimethoxy-9H-carbazol-9-yl)ethyl]phosphonic acid (MeO-2PACz) were purchased from TCI Co., Ltd. Methylammonium iodide (99%), cesium bromide (CsBr , 99.99%), *n*-butylammonium bromide, Fullerene- C_{60} and bathocuproine (BCP, 99%) were purchased from Xi'an Yuri Solar Co., Ltd. Different halogen-substituted PD hydrochloride salts were purchased from Bide Pharmatech Ltd. Anhydrous ethanol was purchased from Sinopharm Chemical Reagent Co., Ltd. *N,N*-dimethylformamide (DMF, anhydrous, 99.8%), *N*-methylpyrrolidone (99.8%), isopropanol (IPA, 99.9%) and chlorobenzene (CB) were purchased from J&K Scientific Co., Ltd. Tetrakis(dimethylamino)tin was purchased from Shanghai Orphant Chemistry Co., Ltd. Indium–tin oxide (ITO) was purchased from Advanced Electron Technology Co., Ltd. Other materials were purchased from Sigma-Aldrich and used as received without any purification.

Synthesis of X-PDI

First, different halogen-substituted piperidine hydrochloride salts were reacted with NaOH to prepare amines. Subsequently, the prepared amines were mixed with hydroiodic acid in a molar ratio of 1:1.1 to synthesize different halogen-substituted PD hydroiodic acid salts (X-PDI). After washing and purifying with ether, the product was collected and dried overnight under high vacuum to obtain a pure white solid.

Preparation of perovskite precursor

The perovskite solution was prepared by dissolving 167.7 mg of formamidinium iodide, 47.7 mg of methylammonium iodide, 605.4 mg of PbI_2 , 82.58 mg of PbBr_2 , 12.5 mg of PbCl_2 , 47.88 mg of CsBr and 2 mg of *n*-butylammonium bromide in a mixture of DMF–*N*-methylpyrrolidone (v/v, 9:1) at a concentration of 1.5 M. For tandem cells, the perovskite concentration was increased to 1.6 M.

Preparation of 2D perovskite film

The 2D thin films were prepared by spin-coating 0.8 M X-PDI and 0.4 M PbI_2 in DMF on fluorine-doped tin oxide (FTO) glass with a ramp of 2,000 rpm (2,000 rpm s^{-1}) and a time of 30 s (annealing 20 min at 100 °C).

Device fabrication of PSCs

Prepatterned ITO glass substrates were ultrasonically cleaned with detergent, deionized water, acetone and IPA for 15 min, followed by treatment with UV-ozone (15 min), and the substrates were transferred into a N_2 glovebox. MeO-2PACz solution (0.5 mg ml^{-1} in anhydrous ethanol) was spin-coated on the ITO substrates at 3,000 rpm for 40 s, followed by thermal annealing at 110 °C for 10 min. The perovskite films were spin-coated at 2,000 rpm for 15 s and 4,000 rpm for 30 s, and then 250 μl CB was dropped onto the spinning substrates 8 s before the end of the spin-coating procedure, followed by thermal annealing at 110 °C for 30 min. Br-PDI and other organic spacers (2 mg ml^{-1} in IPA) were dynamically spin-coated onto the perovskite surface at 5,000 rpm for 30 s and then annealed at 100 °C for 5 min. Then, C_{60} (35 nm) and BCP (10 nm) layers were sequentially thermally deposited on the substrate under high vacuum. Moreover, 100 nm of Ag was thermally evaporated as the metal electrode. Finally, 100 nm of MgF_2 was evaporated onto the glass side as an antireflection layer.

Device fabrication of perovskite/silicon tandem solar cells

For silicon heterojunction bottom cells, 110- μm -thick industrial Czochralski *n*-type silicon wafers were used, with submicron double-sided texturing achieved by means of alkaline soaking.

Following a standard wafer-cleaning process to remove the native oxide, 5–6 nm of intrinsic hydrogenated amorphous silicon layers were deposited on both sides with plasma-enhanced chemical vapour deposition. Phosphorus-doped nanocrystalline silicon oxide and boron-doped nanocrystalline silicon layers were then deposited on the front and rear surfaces, respectively. ITO (80 nm) and indium zinc oxide (IZO; 10 nm) were sputtered on both sides, serving as the rear electrode and recombination layer. Devices were patterned into $1.1 \times 1.1 \text{ cm}^2$ using a metal mask and annealed at 180 °C for 10 min under 20-sun illumination to recover sputtering damage. Ag was thermally evaporated onto the rear side, and the wafers were laser-cut into $2.5 \times 2.5 \text{ cm}^2$ cells for tandem integration. For the perovskite top cells, substrates were treated with UV-ozone for 5 min and immediately transferred into a N_2 -filled glovebox. MeO-2PACz solution (1 mg ml^{-1} in ethanol) was spin-coated onto the IZO layer at 3,000 rpm for 30 s and annealed at 100 °C for 10 min. Then, 1.6 M perovskite solution was spin-coated in a two-step process: 1,000 rpm for 10 s (100 rpm s^{-1}) and then 6,000 rpm for 30 s (1,000 rpm s^{-1}). At 8 s before the end of the programme, 300 μl of CB was quickly dripped on the film centre. After cooling, Br-PDI solution (2 mg ml^{-1} in IPA) was dynamically spin-coated and annealed as described previously. Then 10 nm of C_{60} was thermally evaporated, followed by 12 nm of SnO_2 deposited via atomic layer deposition at 85 °C using tetrakis(dimethylamino)tin(IV) and H_2O as precursors. Dosing and purging times were 0.5 s and 5 s, respectively, for each precursor, with 100 cycles performed. The top IZO electrode (50 nm) was deposited by direct current sputtering in an Ar/O_2 (6 vol%) atmosphere at 2 kW using a rotating target (ϕ , 150 mm; L , 750 mm). Finally, patterned Ag (1 μm) and LiF (110 nm) were thermally evaporated to complete the device.

Material characterization

Ultraviolet–visible (UV–vis) spectra were obtained with a Cary-60 UV–vis spectrophotometer. XRD patterns were acquired with a Shimadzu XRD-6100 diffractometer $\text{Cu K}\alpha$. Steady-state and time-resolved PL were performed using a Spectrofluorometer FLS1000 (Edinburgh Instruments Ltd.) with excitation at 450 nm. PL mappings were conducted utilizing a confocal Raman microscope (inVia Qontor system). PLQY was measured using LuminQY (Ideaoptics) excited at 520 nm in air. Scanning electron microscopy (SEM) images were collected on a JSM-7800F. Atomic force microscopy and KPFM measurements were taken using MFP-3D. UPS and XPS were measured with an AXIS Ultra DLD machine with an incident He source of 21.2 eV. GIWAXS was performed at the BL02U2 beamline of the Shanghai Synchrotron Radiation Facility under a beam wavelength of 0.12398 nm. For cross-sectional measurements, the devices were prepared using a focused ion beam (FIB, TESCAN GAIA3), with protective Pt/carbon layers deposited on the Ag electrode to prevent milling damage. The prepared samples were mounted on a Cu grid and then transferred to the transmission electron microscopy system (TALOS F200X G2) for imaging. For scanning transmission electron microscopy characterization, the concentration of Br-PDI was increased to 8 mg ml^{-1} to facilitate observation of the 2D perovskite phase.

Device characterization

J–*V* curves of single-junction devices were obtained using a xenon-lamp solar simulator (SS-F5-3A, Enli Technology Co. Ltd) under AM 1.5 G conditions (1 sun, 100 mW cm^{-2}) using a Keithley 2401 source metre. For tandem solar cells, an LED-solar simulator was used (G2V, Pico). The EQE spectra of single-junction PSCs were measured using an Enlitech QE-3011 system. The EQE spectra of tandem cells were measured using an Enlitech QE-RX system. Transient photovoltage (TPV) measurements were performed using a Zahner PP211 and Zahner Zennium with a microsecond pulse white light source. Mott–Schottky measurements were also measured on the Zahner with a frequency of 10 kHz. Electrochemical impedance spectra were also measured on the

Zahner in the dark at a bias of 0.8 V by using a frequency from 0.01 Hz to 100 KHz. EQE_{EL} measurements were conducted on an Ocean Optics fibre spectrometer (QE 65 Pro, with an integration sphere) equipped with a Keithley 2400 source metre. The V_{oc} loss caused by non-radiative recombination ($\Delta V_{\text{OC,nrad}}$) can be calculated with the following formula: $\Delta V_{\text{OC,nrad}} = -(k_{\text{B}}T/q)\ln(\text{EQE}_{\text{EL}})$, where k_{B} is the Boltzmann constant, T is temperature and q is the elementary charge. The operational stability of single-junction PSCs with encapsulation was tracked at the MPP under continuous light soaking (100 mW cm⁻², white LED, Bamboo Technology) at around 25 °C in a N₂ atmosphere. The operational stability of tandem solar cells without any encapsulation was tracked at the MPP under continuous light soaking (100 mW cm⁻², white LED, Puri Technology) in a N₂ atmosphere, with a cooling system to maintain a temperature of around 25 °C.

Computational method

The simulated models of PDI-based 2D perovskite materials were created based on the Br-PDI model refined from single-crystal XRD analysis. The interlayer distance was modified in the H, F and Cl-PDI models to match their 020 peaks of powder X-ray diffraction patterns. The Vienna Ab initio Simulation Package (VASP) was used to implement all DFT simulations, and the projector-augmented wave method³⁵ and the DFT-D3 correction method³⁶ were adopted to describe the ion–electron and long-range dispersion interactions, respectively. The generalized gradient approximation level of the functional Perdew–Burke–Ernzerhof exchange correlation was chosen for all calculations³⁷. The cutoff energy was 500 eV to ensure accuracy. K -point sampling was gamma-centred with a $2 \times 3 \times 1$ grid. The convergence criteria of geometrical optimization and electronic minimization were 0.02 eV Å⁻¹ and 1×10^{-5} eV, respectively. After the atomic positions and cell parameters were fully relaxed, slab models with two PD–PbI₄–PD layers were created based on the optimized structures. A total length of 55 Å of the c parameter of the cells was set to avoid interactions between adjacent cells. All slab models were further optimized at the gamma point with PD cations fixed at the bottom. One iodine atom on the top layer was removed in all four slab models, which were again optimized to obtain stable configurations of I vacancy models. To simulate the interactions between perovskite adsorbers and electron transport layers, a C₆₀ molecule was placed on top of the slab models. Full optimization on the atomic positions was implemented, and a denser k mesh of $4 \times 6 \times 1$ was used to calculate the electron density of the complex. Dipole corrections of the periodic systems were adopted to obtain accurate forces and energies in all slab models.

The vacancy formation energy ($E_{\text{f}}^{\text{vacancy}}$) was calculated as follows:

$$E_{\text{f}}^{\text{vacancy}} = E_{\text{vacancy}} - E_{\text{perfect}} + \mu_{\text{element}} \quad (1)$$

where E_{vacancy} , E_{perfect} and μ_{element} are the DFT calculated energy of models with vacancies, perfect models and the chemical potentials of vacant elements, respectively.

Data availability

All data supporting the findings of this study are available in the main text or the Supplementary Information.

References

35. Kresse, G. & Joubert, D. From ultrasoft pseudopotentials to the projector augmented-wave method. *Phys. Rev. B* **59**, 1758–1775 (1999).

36. Grimme, S., Antony, J., Ehrlich, S. & Krieg, H. A consistent and accurate ab initio parametrization of density functional dispersion correction (DFT-D) for the 94 elements H–Pu. *J. Chem. Phys.* **132**, 154104 (2010).
37. Perdew, J. P., Burke, K. & Ernzerhof, M. Generalized gradient approximation made simple. *Phys. Rev. Lett.* **78**, 1396 (1997).

Acknowledgements

We acknowledge BLO2U2 beamline of the Shanghai Synchrotron Radiation Facility (SSRF) for assistance with GIWAXS measurements and the Instrumental Analysis Center of Shanghai Jiao Tong University and the School of Environmental Science and Engineering for assistance with characterizations. We thank H. Yang, T. Zhu and S. Hong from Shanghai Ideaoptics Co., Ltd. for kind assistance in PLQY measurements, and B. Zhu from the National Center of Inspection on Solar Photovoltaic Products Quality for helpful discussion of tandem-cell measurements.

Author contributions

Y.Z., Z.L., Yanming Wang and B.L. directed and supervised the project. Y.Z. conceptualized the research idea and designed the experimental framework. J.G., M.F., H.W. and Z.J. prepared and characterized the films and devices. F.L. contributed to synthesizing the X-PDI spacers and characterizing the perovskite films. B.L. characterized the PLQY. Z.Z., Y. Chang and Yanming Wang conducted DFT calculation. W.Z. conducted the EQE_{EL} measurements. J.G. and K.J. contributed to the TEM measurements. J.G., B.L., Z.Z., X.W., Y. Chen, Yao Wang, Y.M. and Y.Z. drafted, revised and finalized the paper. All the authors revised the paper.

Funding

This work was financially supported by the National Natural Science Foundation of China (NSFC) under grant nos. 22220102002 (Y.Z.), 52203334 (Y.M.), 22522903 (Y. Chen), 52403330 (Yao Wang) and 22409130 (B.L.); the Shanghai Science and Technology Innovation Action Plan under grant no. 24DZ3001203 (B.L.); the Shanghai Science and Technology Commission Program under grant no. 25DZ3001902 (Y.M.); and the Shanghai Jiao Tong University 2030 Initiative under grant no. WH510363004/003 (Y.M.).

Competing interests

Y.Z., J.G., B.L., Yao Wang, Y.M. and Y. Chen are inventors on a patent application (CN202511692493.4) related to this work submitted by Shanghai Jiao Tong University. The other authors declare no competing interests.

Additional information

Supplementary information The online version contains supplementary material available at <https://doi.org/10.1038/s41566-026-01979-z>.

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Peer review information *Nature Photonics* thanks the anonymous reviewers for their contribution to the peer review of this work.

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